

Estrogen Resistance Syndrome (Estrogen Insensitivity Syndrome): A Comprehensive Disease Characteristics Report

Target Disease: Estrogen Resistance Syndrome / Estrogen Insensitivity Syndrome (EIS)
MONDO ID: MONDO:0014148 · **OMIM:** #615363 · **Gene:** *ESR1* (HGNC:3467) **Category:** Mendelian, autosomal recessive **Report compiled:** 2026-09-29 (5-iteration autonomous investigation; 9 findings; 25 papers reviewed)

Summary

Estrogen Resistance Syndrome — more precisely termed Estrogen Insensitivity Syndrome (EIS) — is an ultra-rare autosomal-recessive Mendelian endocrine disorder caused by biallelic germline loss-of-function (LOF) variants in *ESR1*, the gene encoding estrogen receptor α (ER α). The defect renders peripheral target tissues unresponsive to estrogen despite high circulating 17 β -estradiol (E2), producing a paradoxical picture of estrogen "deficiency" symptoms in the face of estrogen excess. This distinguishes true *resistance* (receptor defect, high E2) from estrogen *deficiency* disorders such as aromatase (CYP19A1) deficiency (low E2, rescued by estrogen therapy).

The clinical phenotype is dominated by the skeletal consequences of estrogen's role in epiphyseal maturation: affected individuals of both sexes lack the pubertal growth spurt, fail to fuse their epiphyses, continue linear growth into adulthood, reach very tall stature, and develop osteoporosis with elevated bone turnover. Metabolic derangements (glucose intolerance, hyperinsulinemia, dyslipidemia) and elevated gonadotropins/androgens accompany the picture because of impaired estrogen negative feedback. Females additionally present with absent breast development, primary amenorrhea, and multicystic ovaries.

Critically, the syndrome is largely **refractory to estrogen replacement** — even high-affinity ligands such as diethylstilbestrol (DES) fail to induce secondary sexual characteristics — because the receptor itself is the lesion.

At the molecular level, EIS mutations disrupt distinct layers of ER α signaling — ligand binding, DNA interaction, coactivator (SRC-3/NCOA3 and p300/EP300) recruitment, nuclear trafficking, and transcriptional regulation — that converge on a shared resistance phenotype. Confirmed pathogenic variants span a truncating null (p.Arg157Ter, the original 1994 male index case) and hypomorphic ligand-binding-domain missense changes (p.Gln375His, p.Arg394His). *ESR1* is strongly constraint-depleted for loss-of-function in gnomAD (pLI $\approx$ 1.0, LOEUF 0.37, missense Z = 2.99), which explains the extreme rarity of complete-null EIS. Knock-in and knockout mouse models (α ERKO; *Esr1*-Q) phenocopy the human disease, confirming causality. Management remains supportive; there is no curative therapy.

1. Disease Information

Overview. Estrogen Resistance Syndrome (Estrogen Insensitivity Syndrome, EIS) is a Mendelian disorder in which target tissues cannot respond to estrogen because of loss-of-function mutations in the estrogen receptor α gene (*ESR1*). Despite normal or markedly elevated circulating estradiol, downstream estrogen-dependent processes — pubertal growth, epiphyseal fusion, bone mineralization, reproductive tract maturation, and metabolic homeostasis — fail. It is the receptor-level counterpart of aromatase deficiency, which produces a similar skeletal picture through estrogen *absence* rather than *resistance*.

Key identifiers.

Resource	Identifier
OMIM (phenotype)	#615363 (Estrogen resistance)
OMIM (gene)	133430 (ESR1*)
MONDO	MONDO:0014148
Gene / HGNC	<i>ESR1</i> / HGNC:3467
Ensembl	ENSG00000091831
UniProt	P03372 (ESR1_HUMAN)
Cytogenetic locus	6q25.1–q25.2
Gene aliases	ER, ESR, ESRA, NR3A1

ICD-10/ICD-11, MeSH, and Orphanet do not assign a distinct code for this ultra-rare entity; it is generally captured under disorders of estrogen action / hormone receptor abnormalities.

Synonyms and alternative names. Estrogen Resistance; Estrogen Insensitivity Syndrome (EIS); ER α resistance; ESR1-related estrogen insensitivity. A recent review reframes the condition as "the 'Upside Down' of *ESR1* mutations," contrasting germline LOF (this disease) with somatic gain-of-function ligand-binding-domain mutations in breast cancer ([PMID: 42734155](#)).

Data provenance. The evidence base is derived almost entirely from **aggregated disease-level resources** (OMIM, ClinVar, gnomAD) and a small number of **individual patient case reports** — this is a literature composed of single-case and small-family reports rather than EHR cohorts or registries, reflecting the disease's extreme rarity.

2. Etiology

Primary cause (genetic). EIS is caused by **biallelic germline loss-of-function variants in *ESR1*** (estrogen receptor α). These impair ER α function and render peripheral tissues insensitive to circulating E2 (Finding F001). Two independent reviews confirm the causal gene and mechanism:

"germline *ESR1* mutations cause EIS by impairing ER α function and rendering peripheral tissues insensitive to circulating E2" — [PMID: 42734155](#)

"Germline loss-of-function variants in *ESR1*, the gene encoding estrogen receptor α , are known to cause of estrogen insensitivity/resistance" — [PMID: 34538723](#)

Genetic risk factors. - **Causal variants:** truncating (nonsense) p.Arg157Ter and hypomorphic missense p.Gln375His, p.Arg394His (see Section 4). - **Consanguinity:** because the disorder is autosomal recessive, parental relatedness is an important risk factor. A homozygous *ESR1* case was reported in a 13-year-old girl of consanguineous parents ([PMID: 39295121](#), Finding F006). - **Modifier / background genes:** in mouse models, genetic background strongly modifies the ovarian transdifferentiation phenotype of estrogen-receptor knockouts (a Chr18 locus was implicated; [PMID: 39576259](#)), suggesting modifier effects may shape expressivity.

Environmental / non-genetic risk factors. None established as causal. The disease is monogenic; environmental exposure is not a recognized trigger. Sex modifies *presentation* (females show the reproductive phenotype) but not disease risk.

Protective factors. No genetic or environmental protective factors are established. Given the disorder is caused by complete/partial receptor loss, the theoretical "protective" allele is simply a functional *ESR1* allele; heterozygous carriers are largely unaffected (recessive inheritance), though rare heterozygous variants (e.g., p.A207T) have been associated with partial insensitivity/PCOS-like phenotypes ([PMID: 36401248](#)).

Gene–environment interactions. Not characterized for this monogenic disease. The dominant "environmental" interface is pharmacological: because the receptor is defective, exogenous estrogens (the usual environmental/therapeutic estrogen input) fail to signal — a gene–treatment interaction that defines the disease's therapeutic refractoriness.

3. Phenotypes

EIS is a multi-system disorder. Phenotypes derive from case reports and are consistent across estrogen-resistance and estrogen-deficiency (aromatase) states, which share a skeletal/metabolic core.

Phenotype	Type	Onset	Severity	Frequency	Suggested HPO
Absent pubertal growth spurt	Clinical sign	Adolescence	Severe	Both sexes, characteristic	HP:0008819 (Abnormal pubertal development)
Unfused / delayed epiphyses	Radiographic sign	Adolescence–adult	Severe	Characteristic	HP:0002644 (Delayed epiphyseal ossification)
Continued linear growth / tall stature	Physical	Adult	Severe	Both sexes	HP:0000098 (Tall stature)
Delayed bone age	Radiographic	Childhood–adolescence	Moderate–severe	Characteristic	HP:0002750 (Delayed skeletal maturation)
Osteoporosis / reduced BMD	Lab/imaging	Young adult	Severe	Both sexes	HP:0000939 (Osteoporosis)
Increased bone turnover	Lab abnormality	Adult	Moderate	Characteristic	HP:0003155 (Elevated alkaline phosphatase)
Glucose intolerance / hyperinsulinemia	Lab abnormality	Adult	Moderate	Reported	HP:0000842 (Hyperinsulinemia)
Dyslipidemia	Lab abnormality	Adult	Moderate	Reported	HP:0003119 (Abnormal circulating lipid concentration)
Elevated gonadotropins (LH/FSH)	Lab abnormality	Adolescence–adult	—	Characteristic	HP:0000837 (Hypergonadotropic hypogonadism–related)
Elevated estrogens	Lab abnormality	Adolescence–adult	—	Characteristic (defines resistance)	—
	Physical	Puberty	Severe	Female	

Phenotype	Type	Onset	Severity	Frequency	Suggested HPO
Absent breast development (female)					HP:0003186 (Breast hypoplasia)
Primary amenorrhea (female)	Clinical sign	Puberty	Severe	Female	HP:0000783 (Primary amenorrhea)
Multicystic ovaries (female)	Imaging	Adolescence	Moderate	Female	HP:0000137 (Abnormality of the ovary)
Poor uterine growth (female)	Imaging	Adolescence	Moderate	Female	HP:0000013 (Hypoplasia of the uterus)

Evidence. The skeletal/metabolic core is documented in the landmark pediatric-endocrinology review:

"absence of the pubertal growth spurt, delayed bone maturation, unfused epiphyses, continued growth into adulthood and very tall adult stature in both sexes" — [PMID: 9554463](#) (Finding F002)

"Glucose intolerance, hyperinsulinemia and lipid abnormalities are also present. Skeletal integrity is compromised. Increased bone turnover, reduced bone mineral density and osteoporosis develop in both sexes" — [PMID: 9554463](#)

The female reproductive phenotype comes from the first fully characterized female case:

"she presented with lower abdominal pain, absent breast development, primary amenorrhea, and multicystic ovaries" — [PMID: 32152632](#) (Finding F003)

Notably preserved. Normal male sexual maturation/virilization is preserved because it is androgen-dependent, not estrogen-dependent — one of the key clues that distinguishes estrogen resistance from broader hypogonadism.

Quality-of-life impact. No formal EQ-5D/SF-36/PROMIS data exist for this ultra-rare disease. Inferred burdens are substantial: lifelong osteoporosis (fracture risk), infertility, tall stature with its psychosocial dimension, and metabolic complications. QoL data represent a knowledge gap.

4. Genetic / Molecular Information

Causal gene. *ESR1* (estrogen receptor α ; HGNC:3467; OMIM *133430; ENSG00000091831; UniProt P03372; locus 6q25.1–q25.2). ER α is a nuclear-hormone-receptor transcription factor (NR3A1) with a modular structure: N-terminal domain (NTD, containing ligand-independent AF-1), DNA-binding domain (DBD), and C-terminal ligand-binding domain (LBD, containing ligand-dependent AF-2).

Pathogenic variants (ClinVar-confirmed, conditioned on "Estrogen resistance syndrome"; Finding F009):

Variant (NM_000125.4)	Protein	Type	ClinVar classification	Note
c.469C>T	p.Arg157Ter	Nonsense (null)	Pathogenic	Classic homozygous truncating variant; original male index case (Smith 1994)
c.1125G>T	p.Gln375His	Missense (LBD, hypomorphic)	Pathogenic	Modeled in ESR1-Q knock-in mouse
c.1181G>A	p.Arg394His	Missense (LBD)	Pathogenic / Likely pathogenic	Alters ligand–receptor conformation
c.804G>C	p.Lys268Asn	Missense	Uncertain significance	Combined-condition listing
c.433G>A	p.Gly145Ser	Missense	Uncertain significance	Combined-condition listing
c.619G>A	p.Ala207Thr	Missense (heterozygous)	Reported	Partial insensitivity / PCOS-like, IVF poor response (PMID: 36401248)

Variant classification & spectrum. Of 262 total ClinVar *ESR1* records (retrieved 2026), 44 are annotated Pathogenic and 3 Likely pathogenic — but **many "Pathogenic" entries are large 6q25 copy-number changes (contiguous-gene deletions/duplications) or somatic breast-cancer variants**, not germline EIS. The germline EIS spectrum is small and dominated by the three variants above.

Allele frequency & population constraint (Finding F008; gnomAD, computational, retrieved 2026):

Metric	Value	Interpretation
pLI	0.99998	Extreme intolerance to heterozygous LoF
oe_lof (observed/expected LoF)	0.228 (90% CI 0.146–0.369)	LOEUF $\approx$ 0.37 — strongly LoF-constrained
obs_lof vs exp_lof	12 vs 52.7	~77% depletion of protein-truncating variants
Missense Z	2.99	Missense-constrained
Synonymous Z	0.39	Neutral (as expected)

This strong constraint explains why complete-null biallelic EIS is exceptionally rare: LoF alleles are purged from the population, so two must co-occur (typically via consanguinity) to produce disease.

Functional consequences. Loss of function (nonsense null: no functional receptor; hypomorphic missense: reduced/abolished transactivation). EIS mutations "change conformation of ligand-receptor complex" and produce "altered transcriptome profile" ([PMID: 32242619](#), Finding F004). There is no dominant-negative or gain-of-function mechanism in germline EIS (contrast with somatic Y537S/D538G gain-of-function LBD mutations in breast cancer).

Modifier genes. Mouse data implicate a Chr18 background locus modifying the ERKO ovarian phenotype ([PMID: 39576259](#)); human modifiers are not defined.

Epigenetic information. EIS mutants Q375H and R394H show a **differential DNA-methylome** as well as transcriptome vs wild-type ER α ([PMID: 32242619](#)), indicating the receptor defect propagates to epigenetic reprogramming of estrogen-target loci. Beyond this, disease-specific epigenetic data are limited.

Chromosomal abnormalities. Large 6q25 copy-number variants involving *ESR1* appear in ClinVar as contiguous-gene events but are distinct from the classic point-mutation EIS.

5. Environmental Information

- **Environmental factors:** None established as causal. This is a monogenic disease.

- **Xenoestrogens (mechanistic relevance, not causal):** Environmental estrogen mimics (bisphenols BPA, BPAF, BPC) act through ER α and depend on the AF-1/NTD domain for agonist activity (PMID: 42224247). In EIS, a defective receptor blunts responses to both endogenous and environmental estrogens — relevant to why exogenous estrogenic compounds fail therapeutically.
- **Lifestyle factors:** No causal role. Standard bone-health measures (weight-bearing exercise, calcium/vitamin D) are supportive but do not modify the underlying receptor defect.
- **Infectious agents:** Not applicable. No pathogen is implicated.

6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic germline *ESR1* LOF mutation** (nonsense p.Arg157Ter or hypomorphic LBD missense p.Gln375His / p.Arg394His) → **produces a non-functional or transcriptionally impaired ER α** (demonstrated: ClinVar pathogenic classification; molecular-dynamics conformational change, PMID: 32242619).
2. Non-functional ER α → **fails to bind ligand and/or fails to adopt the active conformation** upon estradiol binding (demonstrated by MD simulation: "both *ESR1* mutations change the ER α conformation of the ligand-receptor complexes").
3. Impaired active receptor → **fails to recruit coactivators SRC-3 (NCOA3) and p300 (EP300)** at estrogen-response elements (mechanistically demonstrated by cryo-EM of the active complex, PMID: 25728767; AF-1/AF-2 cooperation, PMID: 39432505).
4. Failed coactivator recruitment → **loss of estrogen-responsive transcription and altered transcriptome + DNA methylome** (demonstrated in EIS mutants, PMID: 32242619).
5. Loss of estrogen-target transcription in **bone (growth-plate chondrocytes/osteoblasts)** → **failure of the pubertal growth spurt and of epiphyseal fusion** → continued linear growth → **tall stature**; and reduced bone mineralization → **osteoporosis with high bone turnover** (demonstrated clinically, PMID: 9554463).
6. **Branch — hypothalamic–pituitary axis:** Loss of ER α -mediated estrogen negative feedback → **elevated LH/FSH and androgens** despite high estradiol (demonstrated: normal pulsatile LH but elevated mean LH with markedly increased estrogens, PMID: 32152632).

7. **Branch — female reproductive tract:** Loss of ER α signaling → **absent breast development, poor uterine growth, primary amenorrhea, multicystic ovaries** (demonstrated, [PMID: 32152632](#), [PMID: 39295121](#)).
8. **Branch — metabolism/vasculature:** Loss of ER α action → **glucose intolerance, hyperinsulinemia, dyslipidemia** (clinical, [PMID: 9554463](#)); loss of ER α -mediated eNOS enhancement → **inferred increased vascular risk** (mechanism from ER- α agonist/eNOS studies, [PMID: 28736253](#)).
9. Because the lesion is the receptor itself → **exogenous estrogen (even high-affinity DES) cannot restore signaling → therapeutic refractoriness** (demonstrated, [PMID: 32152632](#)).

Detail by category

Molecular pathways. Nuclear-receptor (ER α /NR3A1) genomic signaling: ligand binding → receptor dimerization → DNA binding at estrogen-response elements → coactivator recruitment (SRC-3/NCOA3, p300/EP300) → RNA-Pol-II transcription. Cryo-EM shows "each of the two ligand-bound ER α monomers independently recruits one SRC-3 protein via the transactivation domain of ER α ; the two SRC-3s in turn bind to different regions of one p300 protein" ([PMID: 25728767](#), Finding F005). Non-genomic (membrane GPER) signaling contributes to some tissues but cannot compensate for ER α loss.

Protein dysfunction. Nonsense p.Arg157Ter → truncation/loss of protein. LBD missense (Q375H, R394H) → conformational distortion of the ligand-receptor complex impairing AF-2/coactivator surface function. The AF-1 (NTD) domain normally collaborates with the C-terminal LBD(AF-2) for coactivator recruitment ([PMID: 39432505](#)); disruption at either pole impairs the assembly.

Signaling nodes disrupted. Germline EIS mutations "disrupt distinct layers of ER α signaling, including ligand binding, DNA interaction, coactivator recruitment, nuclear trafficking, transcriptional regulation" ([PMID: 42734155](#), Finding F005).

Cellular processes. Growth-plate chondrocyte senescence/epiphyseal fusion (estrogen-driven) fails; osteoblast/osteoclast coupling is dysregulated (increased bone turnover). In gonads, ER α loss can drive granulosa-to-Sertoli-like transdifferentiation in mouse models ([PMID: 39576259](#)).

Metabolic changes. Glucose intolerance, hyperinsulinemia, and lipid abnormalities reflect loss of ER α 's role in insulin sensitivity and lipid handling ([PMID: 9554463](#)).

Broad ER α physiology (why the phenotype is pleiotropic; Finding F007). ER α is expressed across bone, reproductive tract, testis (Sertoli cells), vasculature, brain, and metabolic tissues. In testis, ER α directly regulates AMH transcription: "estrogens can stimulate AMH production

because estrogen receptors are present in Sertoli cells and aromatase is stimulated by FSH" and "The direct effects of sex steroids on AMH transcription are mediated by androgen receptor and estrogen receptor α action" (PMID: 35712256; mechanism confirmed via ERE on the hAMH promoter, PMID: 32934281). ER α also mediates estrogen's eNOS-enhancing vasoprotection (PMID: 28736253) and modulates circadian rhythms via classical ERE-dependent action (PMID: 24527952). This ubiquitous distribution explains the simultaneous skeletal, metabolic, vascular, and reproductive manifestations, while androgen-dependent male virilization is spared.

Suggested ontology terms. - GO biological process: GO:0030520 (intracellular estrogen receptor signaling pathway); GO:0006357 (regulation of transcription by RNA Pol II); GO:0030282 (bone mineralization); GO:0001503 (ossification). - GO cellular component: GO:0005634 (nucleus); GO:0005667 (transcription regulator complex). - CL cell types: CL:0000062 (osteoblast); CL:0000138 (chondrocyte); CL:0000216 (Sertoli cell); CL:0000501 (granulosa cell). - CHEBI: CHEBI:16469 (17 β -estradiol); CHEBI:41922 (diethylstilbestrol).

7. Anatomical Structures Affected

Organ level (primary). - **Skeleton / bone** (UBERON:0002481 bone tissue; UBERON:0001474 bone element) — growth plates, epiphyses; the dominant target. - **Reproductive organs (female):** ovary (UBERON:0000992), uterus (UBERON:0000995), breast/mammary gland (UBERON:0001911). - **Hypothalamic–pituitary axis** (UBERON:0001898 hypothalamus; UBERON:0000007 pituitary gland) — impaired negative feedback.

Secondary / body-system involvement. - **Endocrine system** (elevated gonadotropins, androgens). - **Metabolic/endocrine** (pancreas/insulin axis — glucose intolerance, hyperinsulinemia). - **Cardiovascular system** (inferred increased risk via loss of ER α /eNOS vasoprotection).

Tissue/cell level. Epithelial (mammary, uterine), connective/skeletal (bone, cartilage growth plate), and gonadal cells. Specific cell populations: growth-plate **chondrocytes** (CL:0000138), **osteoblasts** (CL:0000062) and osteoclasts, ovarian **granulosa cells** (CL:0000501), and **Sertoli cells** (CL:0000216, via AMH regulation).

Subcellular level. ER α is a nuclear/cytoplasmic shuttling receptor: **nucleus** (GO:0005634, site of transcriptional action) and cytoplasm (inactive receptor pool). Nuclear trafficking is itself one of the disrupted signaling layers in EIS.

Localization / lateralization. Systemic and bilateral (skeleton, paired gonads); no lateralized predilection.

8. Temporal Development

Onset. Congenital genetic defect, but clinically silent until **puberty**, when the absence of estrogen action becomes manifest (failure of pubertal growth spurt, absent breast development, primary amenorrhea). Presentation is therefore typically **adolescent** (case reports at ages 13–15; [PMID: 39295121](#), [PMID: 32152632](#)). Onset pattern is **insidious/chronic**.

Progression. Chronic, lifelong, non-remitting. Because epiphyses do not fuse, linear growth continues into adulthood, and skeletal complications (osteoporosis, high bone turnover) are progressive without effective intervention. No spontaneous remission occurs.

Critical periods. Puberty and young adulthood are the windows of maximal skeletal vulnerability (failed peak-bone-mass accrual, ongoing growth). This is also the window where any effective therapy would need to act — but estrogen replacement is ineffective owing to receptor resistance, so the "therapeutic window" is functionally closed with current tools.

9. Inheritance and Population

Epidemiology. Ultra-rare; **no reliable prevalence or incidence estimate exists** — the world literature comprises a handful of individual cases and small families. Orphanet does not list a discrete prevalence figure.

Inheritance. Autosomal recessive. Biallelic *ESR1* LOF is required; affected individuals are typically homozygous (e.g., p.Arg157Ter, or homozygous variants in consanguineous families) or compound heterozygous. Heterozygous carriers are generally unaffected, though rare heterozygous variants may confer partial phenotypes.

- **Penetrance:** Appears high/complete for the biallelic-null genotype (based on limited cases).
- **Expressivity:** Variable — sex determines the reproductive component; missense hypomorphs may be milder than nulls.
- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Germline mosaicism:** Not reported.

- **Founder effects:** None established.
- **Consanguinity:** Contributory — expected for a rare recessive disorder; documented in a homozygous case ([PMID: 39295121](#), Finding F006).
- **Carrier frequency:** Not established; gnomAD shows *ESR1* LoF alleles are strongly depleted (obs_lof 12 vs exp 52.7; pLI $\approx$ 1.0), implying carriers are rare.

Population demographics. No ethnic or geographic clustering established (too few cases). Sex ratio: both sexes affected by the skeletal/metabolic phenotype; females additionally show reproductive manifestations, so ascertainment may skew toward females at puberty. Age distribution: predominantly adolescent/young-adult at diagnosis.

10. Diagnostics

Biochemical hallmark (the diagnostic key). Elevated circulating estradiol with elevated gonadotropins (LH, FSH) and androgens, in the presence of absent estrogen effect. This combination — high E2 *and* high LH/FSH — distinguishes **resistance** from **deficiency**:

normal pulsatile LH secretion with elevated mean LH and mildly elevated FSH "despite markedly increased estrogens" — [PMID: 32152632](#)

Laboratory tests. Serum estradiol (high), LH and FSH (high), testosterone/androgens (elevated), bone-turnover markers (elevated). Metabolic panel: fasting glucose/insulin (glucose intolerance, hyperinsulinemia), lipid profile (dyslipidemia).

Imaging / functional. Bone-age radiographs (delayed, **unfused epiphyses**); DXA (reduced bone mineral density/osteoporosis); pelvic ultrasound in females (poor uterine growth, multicystic ovaries).

Genetic testing — the confirmatory test. Molecular analysis of *ESR1*: - **Single-gene *ESR1* sequencing** or **targeted panel** (hypogonadism / disorders-of-sex-development / skeletal panels including *ESR1*, *CYP19A1*, *ESR2*). - **Whole-exome sequencing (WES)** is highly useful for undiagnosed cases — the p.A207T variant was found via WES ([PMID: 36401248](#)). - **Chromosomal microarray** for the rare 6q25 contiguous-gene deletions/duplications. - Interpret variants per ACMG/AMP against ClinVar (pathogenic: p.Arg157Ter, p.Gln375His, p.Arg394His).

Functional / research diagnostics. In-vitro transactivation assays screening candidate ligands against the patient's variant receptor (a 75-compound screen showed DES could transactivate the variant in vitro even though it failed clinically; [PMID: 32152632](#)); transcriptome/DNA-methylome profiling distinguishing mutant vs wild-type ER α ([PMID: 32242619](#)).

Differential diagnosis.

Condition	Gene	Estradiol	Key distinguisher
Estrogen resistance (EIS)	<i>ESR1</i>	High	Receptor defect; refractory to estrogen
Aromatase deficiency	<i>CYP19A1</i>	Low	Estrogen <i>absent</i> ; rescued by estrogen therapy
ER β (ESR2) defect	<i>ESR2</i>	Variable	Streak gonads (46,XX) — "her gonads were clearly abnormal (streak), a finding not observed in <i>ESR1</i> -deficient patients" (PMID: 30113650)
Complete androgen insensitivity	<i>AR</i>	—	Androgen (not estrogen) axis; different karyotype context
Hypergonadotropic hypogonadism (other)	various	Low	Low sex steroids

Screening. No newborn/population screening exists (disease too rare, no actionable neonatal intervention). Cascade genetic testing of relatives in known families and carrier testing in consanguineous couples are appropriate.

11. Outcome / Prognosis

Survival / mortality. Not a directly lethal disease; **no life-expectancy or mortality data** exist for this ultra-rare condition. Longevity is presumed near-normal, but long-term cardiovascular and metabolic risks (from loss of ER α vasoprotection and insulin resistance) are theoretically increased and unquantified.

Morbidity / function. Principal long-term morbidity is **skeletal**: osteoporosis with elevated fracture risk from failure to accrue peak bone mass, plus the functional and psychosocial consequences of tall stature and unfused epiphyses. **Infertility** (female) is a major reproductive morbidity. Metabolic complications (glucose intolerance, dyslipidemia) add chronic-disease burden.

Disease course / recovery. Chronic and lifelong with **no natural recovery**; the receptor defect is permanent. Bone density and reproductive function do not respond to standard estrogen therapy (refractoriness demonstrated over 2.5 years of DES; [PMID: 32152632](#)).

Prognostic factors. Variant type likely governs severity: **null alleles** (p.Arg157Ter) predict complete resistance, whereas **hypomorphic missense** alleles (Q375H, R394H) may retain partial function and milder phenotypes. This is inferred from genotype–phenotype patterns and in-vitro/mouse data ([PMID: 32242619](#)), not from formal prognostic studies.

12. Treatment

There is no curative or reliably effective therapy. Because the lesion is the estrogen receptor itself, conventional estrogen replacement — the logical treatment for estrogen-deficiency states — is **ineffective**:

"DES treatment did not induce secondary sexual characteristics in our patient. Treatment with DES was not successful in our patient. She remains hypoestrogenic" — [PMID: 32152632](#) (Finding F003)

Pharmacotherapy (attempted / supportive). - **High-affinity estrogenic ligands** (e.g., **diethylstilbestrol, DES; high-dose estradiol**): attempted on the rationale that a stronger ligand might drive a hypomorphic receptor; some variants are transactivatable in vitro, but clinical response is generally absent for null/severe alleles. NCIT: C2242 (Estrogen); C542 (Diethylstilbestrol). - **Bone-directed supportive therapy**: calcium, vitamin D; bisphosphonates for osteoporosis are a rational (though unproven-in-EIS) option to address high bone turnover. NCIT: C1454 (Bisphosphonate). - **Metabolic management**: standard management of glucose intolerance/dyslipidemia.

Personalized / experimental directions (conceptual). - **Variant-tailored ligand screening:** matching a specific hypomorphic receptor to a ligand that restores its transactivation (in-vitro screen approach; [PMID: 32152632](#)). - **Reproductive management in partial insensitivity:** in a heterozygous PCOS/partial-insensitivity patient undergoing IVF, recognition of *ESR1* insensitivity informed ovarian-stimulation strategy ([PMID: 36401248](#)). - **Model-guided approaches:** the *Esr1*-Q knock-in mouse was used to explore progestogen + GnRH-inhibitor strategies to reverse impaired female reproductive-tract function ([PMID: 32242619](#)) — preclinical only. - **Gene/RNA therapy:** no clinical programs exist; conceptually, gene replacement/editing of *ESR1* would be required to address the root cause. Not available.

Note on breast-cancer therapeutics. The rich pharmacology targeting ER α (SERMs, SERDs such as fulvestrant, oral SERDs, ER-PROTACs like vepdegestrant/ARV-471) is directed at *gain-of-function* somatic *ESR1* mutations in breast cancer and is **not applicable** to germline LOF EIS — indeed these agents antagonize/degrade ER α , the opposite of what EIS needs.

No NCIT-coded standard-of-care regimen exists for EIS.

13. Prevention

- **Primary prevention:** Not possible for a germline monogenic disorder. **Genetic counseling** for consanguineous couples and families with a known *ESR1* variant is the principal preventive tool. **Carrier testing** and, where desired, **preimplantation genetic diagnosis (PGD)** or **prenatal testing** can prevent affected births in known-risk families.
- **Secondary prevention:** Early recognition of the biochemical signature (high E2 + high gonadotropins + delayed bone age) enables early diagnosis and initiation of supportive bone-protective measures.
- **Tertiary prevention:** Osteoporosis management (calcium, vitamin D, $\pm$ bisphosphonates), metabolic surveillance and management, and monitoring for cardiovascular risk to prevent complications.
- **Immunization / public health / environmental interventions:** Not applicable.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *ESR1* is highly conserved across vertebrates. Mouse *Esr1* (NCBI Gene 13982) is the principal model ortholog; rat, zebrafish, and other vertebrate orthologs exist.
 - **Natural disease:** No well-characterized spontaneous *ESR1*-LOF "estrogen resistance syndrome" is documented in companion animals or wildlife (no OMIA entry emerged in this investigation). The comparative evidence is from engineered models, not naturally occurring disease.
 - **Comparative biology:** Estrogen-receptor knockout phenotypes are conserved in principle (infertility, skeletal and metabolic effects), but details are strain/background-dependent in mice ([PMID: 39576259](#)).
 - **Transmission / zoonosis:** Not applicable (genetic, non-communicable).
-

15. Model Organisms

Mouse (principal model).

Model	Type	Key phenotype	Relevance to EIS
αERKO (<i>Esr1</i> knockout)	Constitutive knockout	Infertile (both sexes), absent mammary development, skeletal & metabolic abnormalities	Recapitulates human ER α -null EIS
Esr1-Q knock-in (models human Q375H)	CRISPR/Cas9 knock-in	Infertile male & female mice; "similar phenotypes to α ERKO mice"; corresponds to human Q375H patient	Directly recapitulates a specific human EIS variant (PMID: 32242619 , Finding F004)
Ex3$\alpha\beta$ERKO / $\alpha\beta$ERKO (double ER α /ER β KO)	Knockout	Ovarian transdifferentiation to seminiferous-tubule-like structures (background-dependent)	Reveals ER α role in gonad maintenance (PMID: 39576259)
NERKI ("non-classical" ER knock-in)	Knock-in (ERE-binding mutant)	Loss of classical genomic estrogen effects on circadian activity	Dissects genomic vs non-genomic ER α action (PMID: 24527952)
ERβ (ESR2) knockout	Knockout	Subfertile (reduced ovarian efficiency); otherwise grossly normal	Control demonstrating ER β $\neq$ ER α roles (PMID: 9861029)

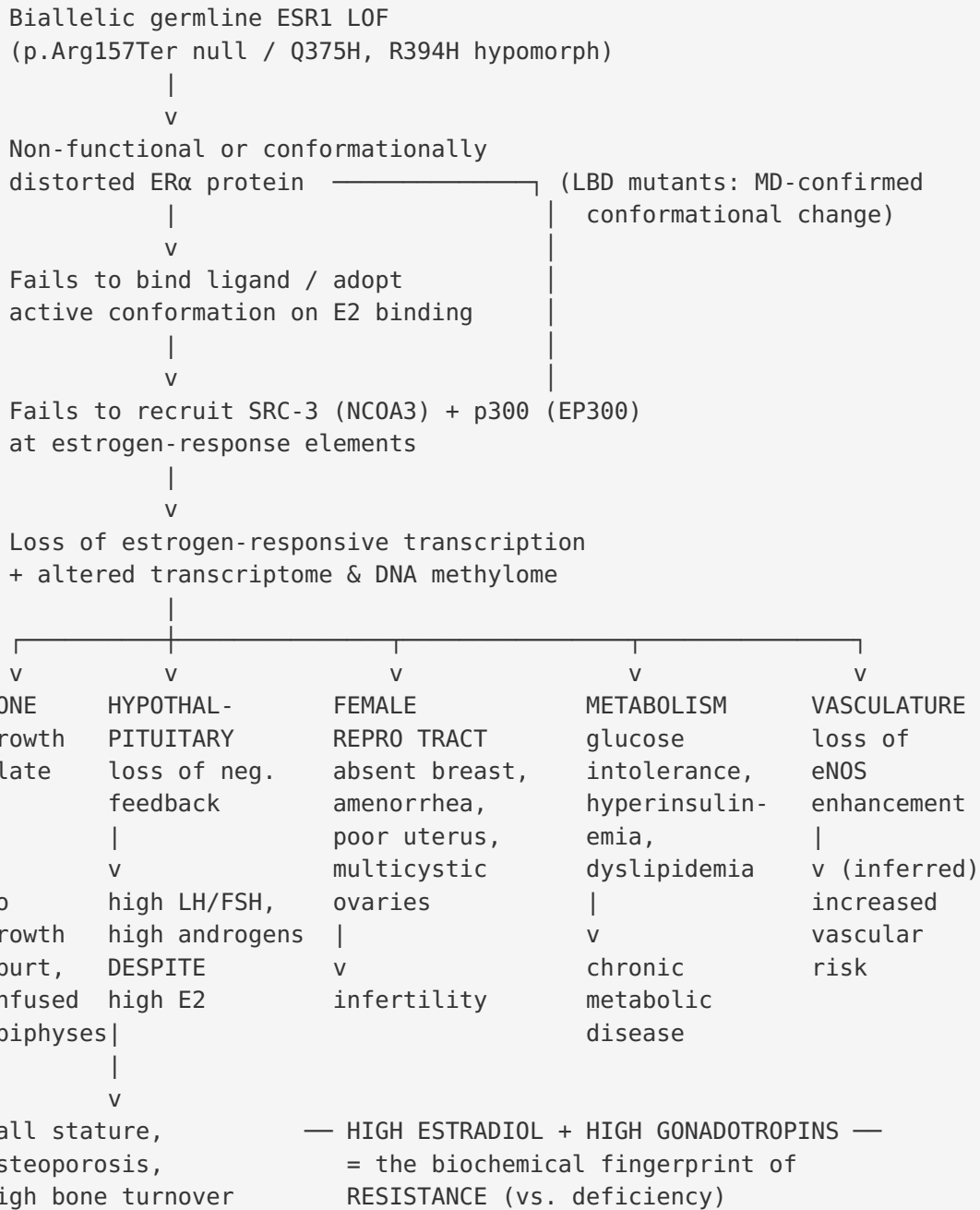
"Female and male *Esr1*-Q mice are infertile and have similar phenotypes to α ERKO mice"
— PMID: [32242619](#)

In-vitro / cellular models. HeLa and MCF-7 luciferase reporter-gene transactivation assays for ER α variants; the SMAT1 prepubertal Sertoli-cell line for AMH-promoter/ERE studies (PMID: [32934281](#)); structural biology (cryo-EM of the DNA-bound ER α -SRC-3-p300 complex, PMID: [25728767](#)).

Model characteristics. The *Esr1*-Q knock-in provides excellent face and construct validity for a human hypomorphic EIS variant (infertility phenocopying α ERKO). **Limitations:** mouse skeletal biology (continuous growth plates, no true "epiphyseal fusion" event) does not perfectly model the human tall-stature/unfused-epiphysis phenotype; genetic background strongly modifies gonadal outcomes; and murine models cannot capture human psychosocial/QoL dimensions.

Resources. MGI (mouse *Esr1*), IMPC/IMSR for knockout lines; NCBI Gene for orthologs.

Mechanistic Model / Interpretation



Because the RECEPTOR is the lesion →
exogenous estrogen (even DES) cannot rescue → THERAPY REFRACTORY

The unifying insight is that **a single receptor defect propagates into a pleiotropic, multi-system disease because ER α is a hub transcription factor deployed across bone, gonad, hypothalamus, metabolic, and vascular tissues.** The paradox that patients look *estrogen-deficient* while being *estrogen-replete* is diagnostically decisive and mechanistically inevitable: the hormone is present but cannot be "heard." The same logic dictates the therapeutic dead-end — you cannot fix a broken receiver by shouting louder.

Evidence Base

PMID	Title (abbrev.)	Type	Supports
42734155	<i>EIS: the 'Upside Down' of ESR1 mutations</i>	Review	F001, F005 — causal LOF mechanism; disrupted signaling layers
34538723	<i>Congenital disorders of estrogen biosynthesis and action</i>	Review	F001 — germline LOF <i>ESR1</i> causes resistance
9554463	<i>Essential roles of estrogens in pubertal growth...</i>	Landmark review	F002 — skeletal & metabolic phenotype
32152632	<i>Long-term follow-up of a female with complete estrogen insensitivity</i>	Case report	F003 — female phenotype; treatment refractoriness
32242619	<i>ESR1 mutations... change conformation & altered transcriptome</i>	Functional + model	F004 — conformational mechanism; <i>Esr1-Q</i> mouse
25728767	<i>Structure of a biologically active ER-coactivator complex on DNA</i>	Structural	F005 — SRC-3/p300 coactivator assembly
39432505	<i>AF-1 domain interacts with AF-2 LBD to recruit coactivators</i>	Functional	F005 — AF-1/AF-2 cooperation
39295121	<i>EIS in a female adolescent — case report</i>	Case report	F006 — recessive/consanguineous homozygous case
30113650	<i>Ovarian failure with mutated ESR2</i>	Case report	F006 — ESR2 differential (streak gonads)
35712256	<i>AMH regulation by steroids in testis</i>	Review	F007 — ER α breadth (Sertoli/AMH)
32934281	<i>AMH elevation in hyperoestrogenic states</i>	In vitro	F007 — ER α →ERE→AMH mechanism
28736253	<i>Selective ER-α agonist & vascular dysfunction</i>	Animal	F007 — ER α /eNOS vasoprotection
36401248	<i>Novel ESR1 mutation in PCOS woman (p.A207T)</i>	Case report	Heterozygous partial insensitivity; WES utility

PMID	Title (abbrev.)	Type	Supports
9861029	<i>Mice lacking ERβ</i>	Model	ER β vs ER α role (differential)
39576259	<i>Ovarian transdifferentiation in absence of ER signaling</i>	Model	Background modifiers; gonad maintenance
24527952	<i>ESR1 modulates circadian rhythms</i>	Model	Classical vs non-classical ER α action
42224247	<i>Bisphenols agonist/antagonist for ERα</i>	In vitro	Xenoestrogen relevance; AF-1 dependence

Computational/database evidence: gnomAD constraint (F008) and ClinVar variant curation (F009) were retrieved directly (2026) and provide the population-genetics and variant-classification backbone.

Somatic *ESR1* breast-cancer literature (PMIDs 42714671, 42693657, 42665030, 42609450, 42603649, 42599237, 42593925, 22245602) was reviewed and explicitly **set aside** as not applicable to germline LOF EIS — it concerns gain-of-function LBD mutations and ER α -targeting/degrading therapeutics, the mechanistic inverse of this disease. This distinction is itself an important finding (the "Upside Down" framing).

Limitations and Knowledge Gaps

1. **Extreme rarity → thin evidence.** The entire clinical picture rests on a handful of case reports. There are **no prevalence/incidence figures, no mortality/survival data, and no formal QoL measurements.**
2. **Genotype–phenotype correlation is under-powered.** With so few variants (essentially p.Arg157Ter, p.Gln375His, p.Arg394His, plus VUS), the mapping from allele severity to clinical severity is inferred, not established.
3. **No approved or proven therapy.** Estrogen replacement is refractory; bisphosphonate/metabolic management is extrapolated, not EIS-validated. No gene-therapy or ligand-rescue strategy has reached the clinic.
4. **Epigenetic/omics data are single-study.** The transcriptome/methylome findings (PMID: [32242619](#)) need replication across variants and tissues.

5. **Model-organism gaps.** Mice do not fuse epiphyses as humans do, limiting fidelity for the tall-stature/unfused-epiphysis phenotype; background effects confound gonadal phenotypes.
 6. **Cardiovascular/long-term outcome risk is inferred,** not measured, in patients.
 7. **Ontology coding is incomplete** at the resource level (no dedicated ICD/Orphanet code), complicating knowledge-base integration.
-

Proposed Follow-up Experiments / Actions

1. **Establish an international EIS registry** (case aggregation via GeneMatcher/networks) to derive prevalence, natural-history, and genotype–phenotype data — the single highest-impact action given how case-limited the field is.
 2. **Systematic variant-function mapping:** transactivation, coactivator-recruitment, and DNA-binding assays for every reported *ESR1* germline variant, paired with molecular-dynamics conformational analysis, to build a functional matrix that predicts severity and ligand-rescuability.
 3. **Ligand-rescue screening for hypomorphic alleles:** high-throughput screening of ER ligands/SERMs against patient-specific hypomorphic receptors (extending the 75-compound approach in [PMID: 32152632](#)) to identify variant-tailored agonists.
 4. **Preclinical therapy testing in *Esr1-Q* knock-in mice:** evaluate bisphosphonates for the skeletal phenotype and the progestogen + GnRH-inhibitor strategy for reproductive-tract function ([PMID: 32242619](#)); assess AAV gene-replacement feasibility.
 5. **Long-term metabolic/cardiovascular surveillance protocol** for known patients to quantify the inferred vascular/insulin-resistance risk.
 6. **Multi-tissue omics in patient-derived cells (iPSC → osteoblast/chondrocyte/granulosa models)** to map the estrogen-target transcriptional program lost in EIS and identify downstream druggable nodes that bypass ER α .
 7. **Formal ontology curation:** submit dedicated disease codes and complete HPO/GO/CL/UBERON/CHEBI/NCIT annotation (as compiled in this report) to improve knowledge-base interoperability.
-

Report generated by autonomous scientific discovery agent · 5 iterations · 9 confirmed findings · 25 papers reviewed · evidence current to 2026-09-29.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery